Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

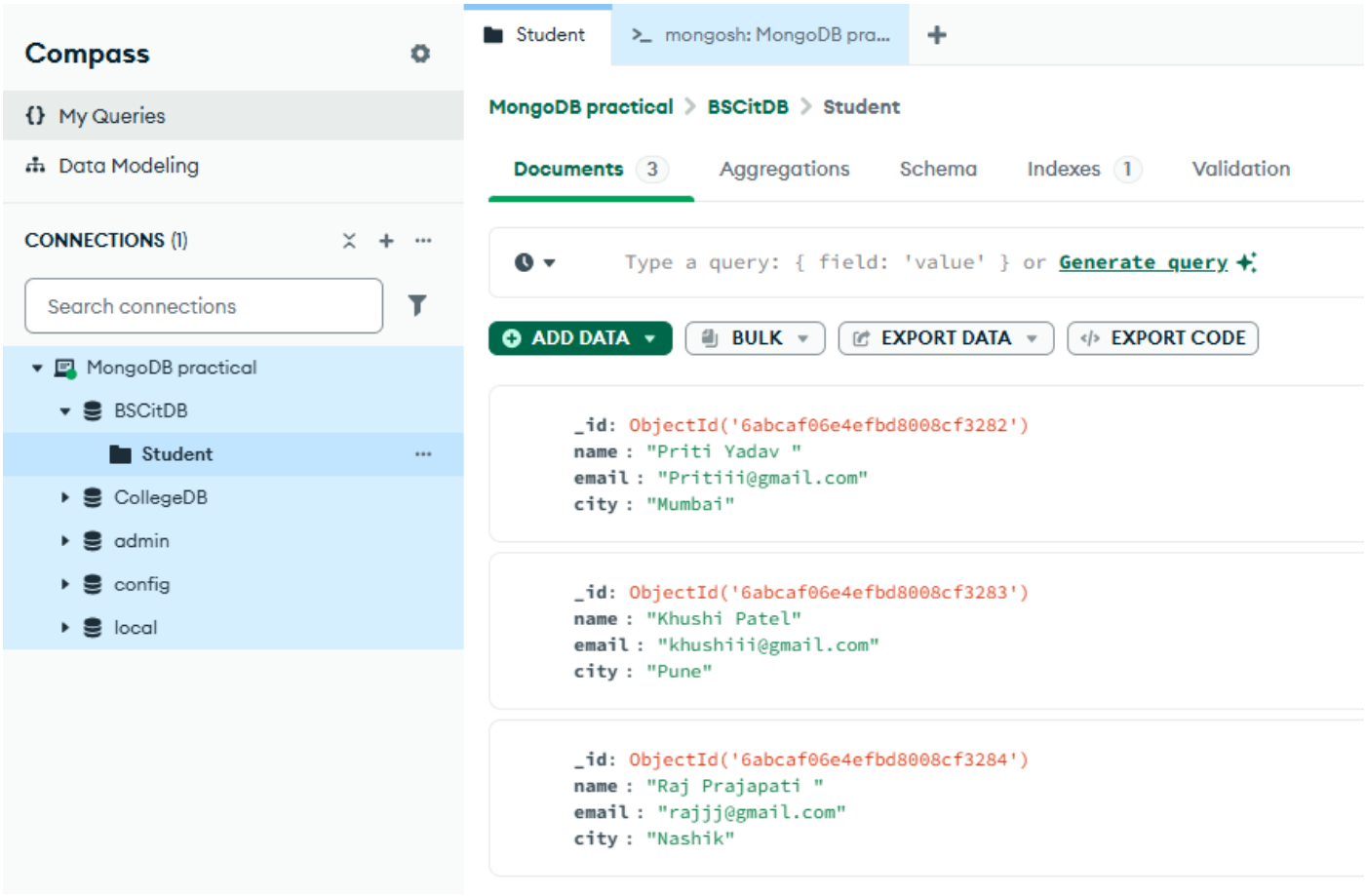
Full Stack Development Using Mern Stack

Practical – IX

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Class: TYIT	Batch: 2
Date of Assignment: 30-09-26	Date/Time of Submission: 30-09-26

9.User Data Management

a. Create an API to add user details using Express and MongoDB.



App.jsx :

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
// Store all users
const [users, setUsers] = useState([]);
// Form data
const [name, setName] = useState("");
const [email, setEmail] = useState("");
const [city, setCity] = useState("");
// Store ID of user being edited
const [editId, setEditId] = useState(null);
// Get Users from MongoDB
```

```
const getUsers = () => {
axios.get("http://localhost:3000/users")
.then((response) => {
setUsers(response.data);  })
.catch((error) => {
console.log(error);
});  };
// Load users when page starts
useEffect(() => {getUsers();}, []);
// Add or Update User
const saveUser = () => { const user = {
name: name,
email: email,
city: city
};
// UPDATE existing user
if (editId) {axios.put(`http://localhost:3000/users/${editId}`, user)
.then((response) => {alert(response.data.message);
clearForm();
getUsers();
})
.catch((error) => {console.log(error);});
}
// ADD new user
else {axios.post("http://localhost:3000/users", user)
.then((response) => {alert(response.data.message);
clearForm();
getUsers();
})
.catch((error) => {console.log(error);});
}
};
// Edit User
const editUser = (user) => {setEditId(user._id);
setName(user.name);
setEmail(user.email);
setCity(user.city);
};
// Delete User
const deleteUser = (id) => {
if (window.confirm("Are you sure you want to delete this user?")) {
axios.delete(`http://localhost:3000/users/${id}`)
.then((response) => {alert(response.data.message);
getUsers();
})
.catch((error) => {console.log(error);});   }};
// Clear Form
const clearForm = () => {
setName("");
setEmail("");
```

```
setCity("");
setEditId(null);   };
return (
<div>
<h1>User Management System</h1>
{/* User Form */}
<h2>
{editId ? "Update User" : "Add User"}
</h2>
<input type="text"
placeholder="Enter Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />
<input type="email"
placeholder="Enter Email"
value={email}
onChange={(e) => setEmail(e.target.value)}
/>
<br /><br />
<input type="text"
placeholder="Enter City"
value={city}
onChange={(e) => setCity(e.target.value)}
/>
<br /><br />
<button onClick={saveUser}>
{editId ? "Update User" : "Add User"}
</button>
{editId && (<button onClick={clearForm}>
Cancel
</button>   )}
<hr />
{/* Display Users */}
<h2>User List</h2>
<table border="1">
<thead>
<tr>
<th>Name</th>
<th>Email</th>
<th>City</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
{users.map((user) => (
<tr key={user._id}>
<td>{user.name}</td>
<td>{user.email}</td>
```

```
<td>{user.city}</td>
<td>
<button onClick={() => editUser(user)}>
Edit
</button>
<button onClick={() => deleteUser(user._id)}>
Delete
</button>
</td>
</tr>
))}
</tbody>
</table>
</div>
);}
export default App;
```

server.js :

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/UserDB")
.then(() => {
console.log("MongoDB Connected Successfully");
})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);
});
// GET - Retrieve Users
app.get("/users", async (req, res) => {
try {
const users = await mongoose.connection.db
.collection("users")
.find()
.toArray();
res.json(users);
}
catch (error) {
res.status(500).json({message: "Error fetching users"});
}
});
// POST - Add User
app.post("/users", async (req, res) => {
```

```
try {
const user = req.body;
await mongoose.connection.db
.collection("users")
.insertOne(user);
res.json({message: "User added successfully"});  }
catch (error) {
res.status(500).json({message: "Error adding user"});  }
});
// PUT - Update User
app.put("/users/:id", async (req, res) => {
try {
const id = new mongoose.Types.ObjectId(req.params.id);
const updatedUser = req.body;
await mongoose.connection.db
.collection("users")
.updateOne({ _id: id },
{
$set: {
name: updatedUser.name,
email: updatedUser.email,
city: updatedUser.city
}
}
);
res.json({message: "User updated successfully"});
}
catch (error) {
res.status(500).json({message: "Error updating user"});
}
});
// DELETE - Delete User
app.delete("/users/:id", async (req, res) => {
try {
const id = new mongoose.Types.ObjectId(req.params.id);
await mongoose.connection.db
.collection("users")
.deleteOne({
_id: id

});
res.json({message: "User deleted successfully"});
}
catch (error) {
res.status(500).json({message: "Error deleting user"});
}
});
// Start Server
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
```

});

b. Design a React form to send user data to the backend.

```
{
  _id: ObjectId('6abcb0a7e9bf7fcc10469bfb'),
  name: 'Priti Yadav',
  email: 'Pritiii@gmail.com',
  city: 'Mumbai'
}
{
  _id: ObjectId('6abcb0a7e9bf7fcc10469bfd'),
  name: 'Raj Prajapati',
  email: 'raj@gmail.com',
  city: 'Nashik'
}
{
  _id: ObjectId('6abcc1bd2357c957d758b27f'),
  name: 'Aashwin ',
  email: 'ashwin@gmail.com',
  city: 'Delhi'
}
{
  _id: ObjectId('6abcc6e9e4efbd8008cf3293'),
  name: 'Ashwin',
  email: 'Ashwinh@gmail.com',
  city: 'Delhi'
}
```

c. Retrieve and display user data from MongoDB.

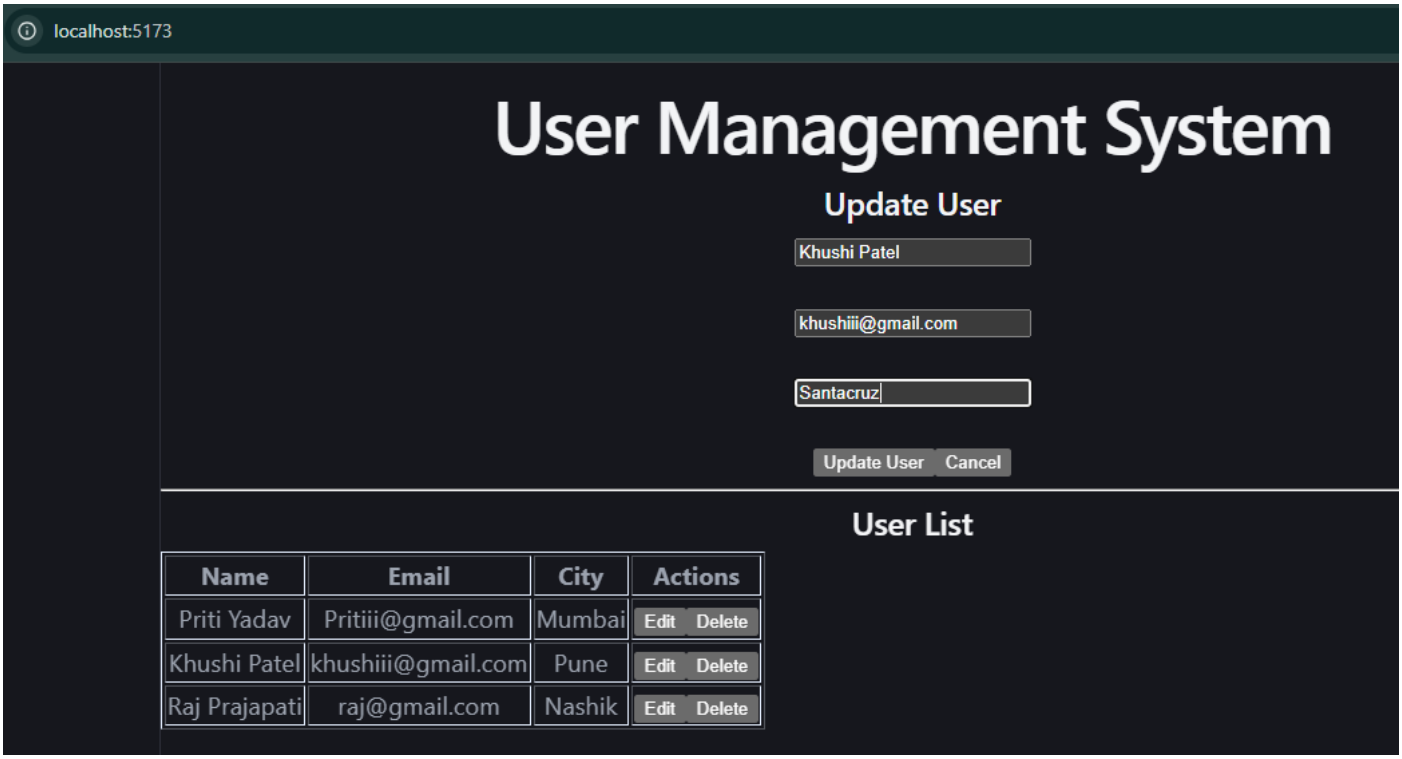
```
_id: ObjectId('6abcb0a7e9bf7fcc10469bfb')
name : "Priti Yadav"
email : "Pritiii@gmail.com"
city : "Mumbai"
```

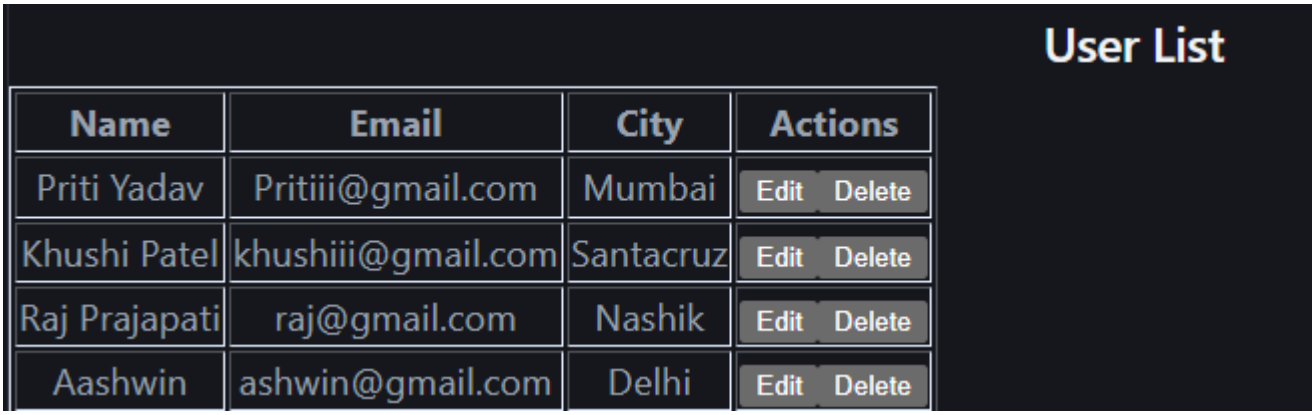
```
_id: ObjectId('6abcb0a7e9bf7fcc10469bfd')
name : "Raj Prajapati"
email : "raj@gmail.com"
city : "Nashik"
```

```
_id: ObjectId('6abcc1bd2357c957d758b27f')
name : "Aashwin "
email : "ashwin@gmail.com"
city : "Delhi"
```

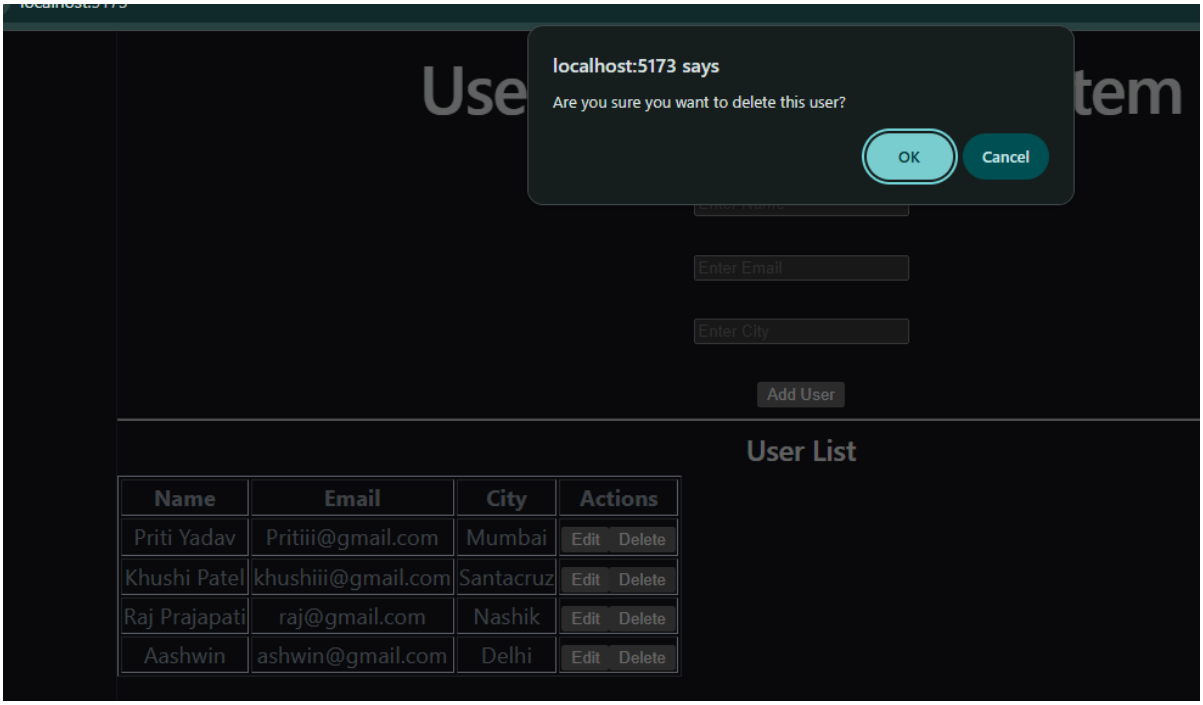
```
_id: ObjectId('6abcc6e9e4efbd8008cf3293')
name : "Ashwin"
email : "Ashwinh@gmail.com"
city : "Delhi"
```

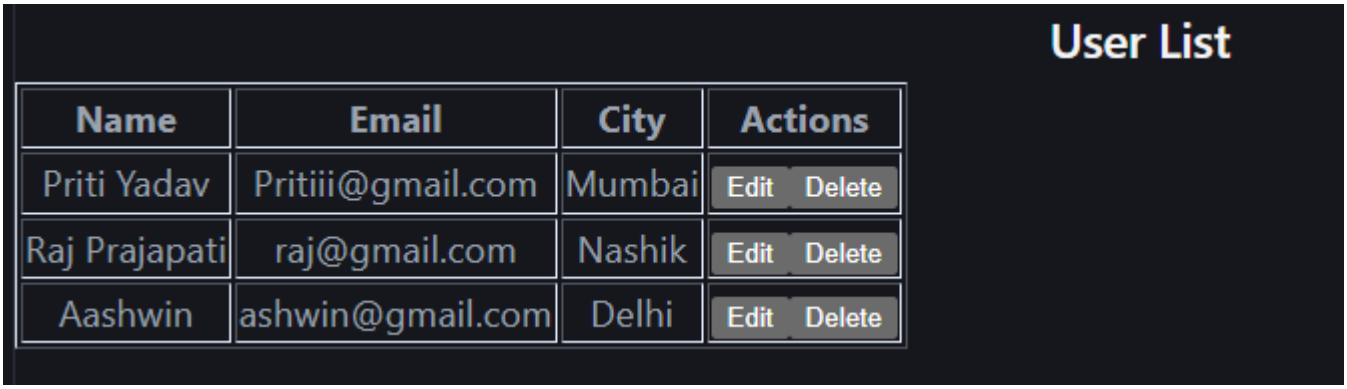
d. Implement an update operation for existing records.



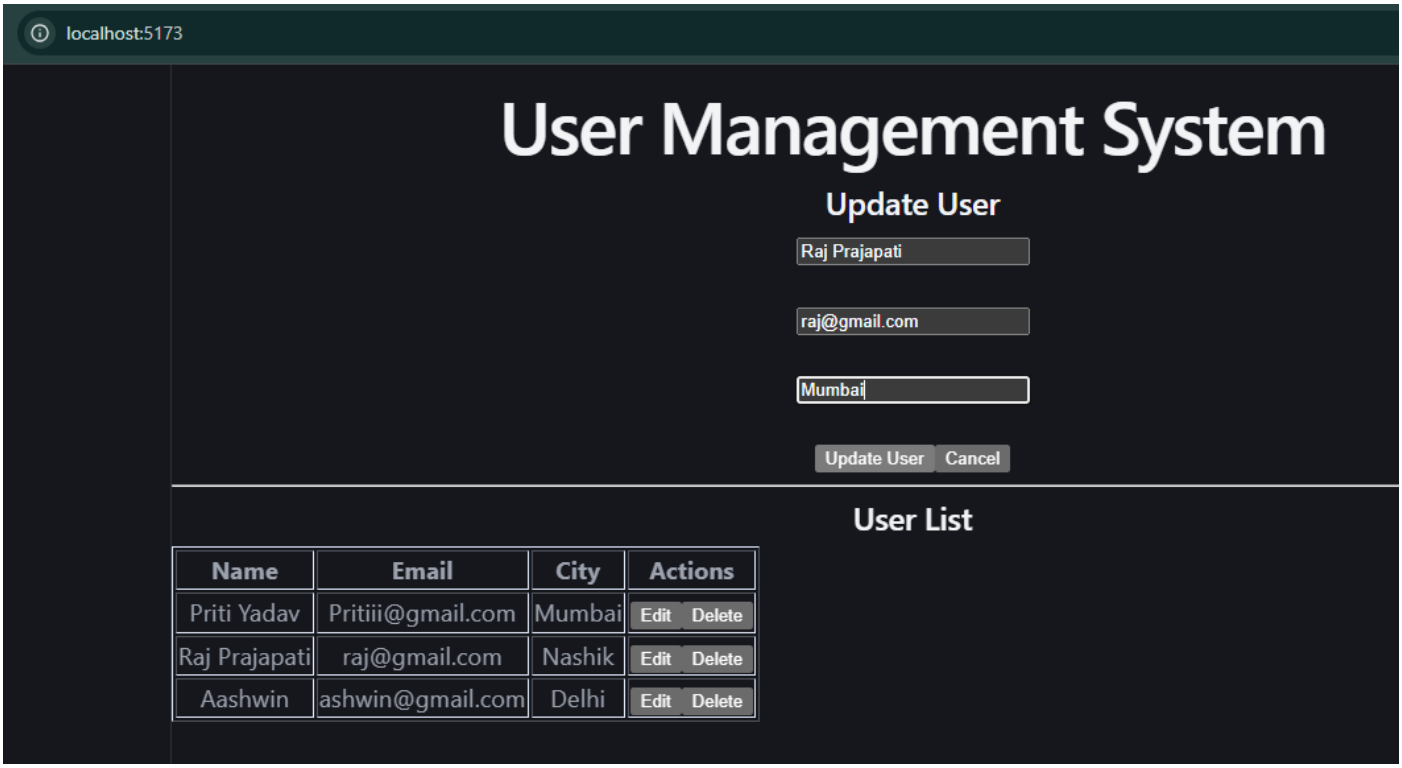


e. Implement the delete operation for selected records.





f. Display updated records dynamically in React.




```
_id: ObjectId('6abcb0a7e9bf7fcc10469bfd')
name : "Raj Prajapati"
email : "raj@gmail.com"
city : "Mumbai"
```